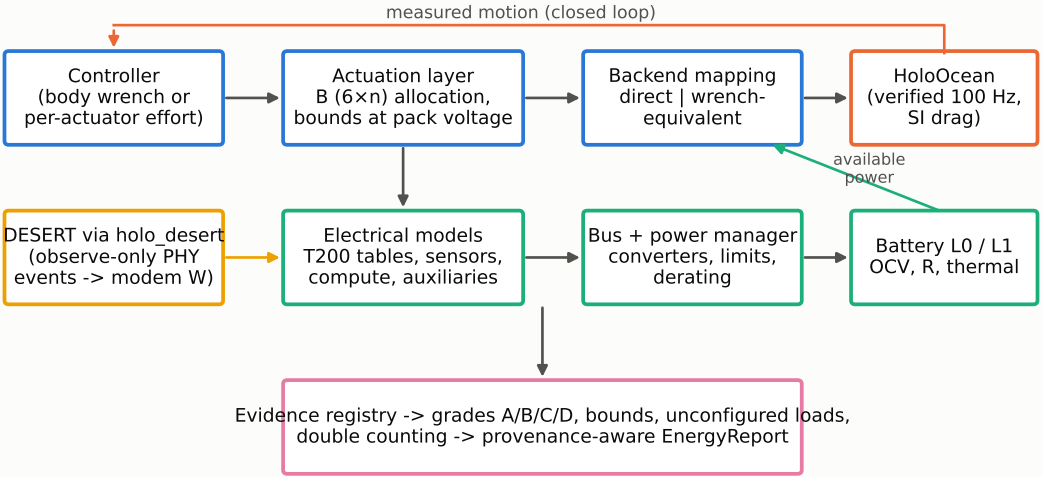
measured motion (closed loop)
Controller
(body wrench or
per-actuator effort)
Actuation layer
B (6×n) allocation,
bounds at pack voltage
Backend mapping
direct | wrench-
equivalent
HoloOcean
(verified 100 Hz,
SI drag)
available
power
DESERT via holo_desert
(observe-only PHY
events -> modem W)
Electrical models
T200 tables, sensors,
compute, auxiliaries
Bus + power manager
converters, limits,
derating
Battery L0 / L1
OCV, R, thermal
Evidence registry -> grades A/B/C/D, bounds, unconfigured loads,
double counting -> provenance-aware EnergyReport